SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -1

CO2_AT2

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Course code : CSA6502

Course name: Generative Ai And Large Scale Models

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Case Study

Develop an appropriate solution for the given Generative AI/LLM scenario. Your answer should include analysis, justification, suitable model selection (GPT/BERT/RLHF/LoRA/PEFT where applicable), implementation approach, expected outcome, and limitations.

Topic: Financial fraud detection

ANSWER :-

1. Introduction

The rapid growth of digital banking, online payments, UPI, credit cards, mobile wallets, and e-commerce has transformed the financial industry by making transactions faster, easier, and more convenient. However, this digital transformation has also increased the risk of financial fraud. Cybercriminals use techniques such as credit card fraud, identity theft, phishing, account takeover, fake loan applications, insurance fraud, and money laundering to exploit weaknesses in financial systems. These fraudulent activities result in significant financial losses, reduced customer trust, and increased operational costs for financial institutions.

Traditional fraud detection systems mainly rely on predefined rules and historical transaction patterns. Although these methods can detect known fraud types, they struggle to identify new and evolving fraud techniques. They also generate a high number of false positives, where legitimate transactions are incorrectly flagged as fraudulent, leading to customer inconvenience and additional manual investigation.

Artificial Intelligence (AI) and Machine Learning (ML) have improved fraud detection by analyzing large volumes of transaction data and identifying hidden patterns. However, many conventional ML models focus mainly on structured numerical data and have limited ability to process unstructured information such as transaction descriptions, customer complaints, emails, and chat messages.

The emergence of Generative AI and Large Language Models (LLMs) has introduced a more advanced approach to financial fraud detection.

2. Problem Statement

The rapid increase in digital banking, online payments, UPI transactions, credit card usage, and e-commerce has led to a significant rise in financial fraud. Fraudsters continuously develop new techniques such as phishing, identity theft, account takeover, fake transactions, and money laundering, making fraud detection more challenging. Traditional rule-based and machine learning systems often fail to identify these evolving fraud patterns because they mainly rely on predefined rules and historical data.

Existing fraud detection methods also produce a high number of false positives, where genuine transactions are incorrectly marked as fraudulent. This increases manual verification efforts, delays transaction processing, reduces customer satisfaction, and increases operational costs. Furthermore, these systems have limited capability to analyze unstructured data such as transaction descriptions, customer complaints, emails, and chat messages, which may contain valuable fraud-related information.

Therefore, there is a need for an intelligent fraud detection system that can analyze both structured and unstructured financial data, detect fraud in real time, adapt to new fraud patterns, and provide explainable predictions. By integrating Generative AI, GPT, BERT, RLHF, LoRA, and PEFT, the proposed system aims to improve fraud detection accuracy, reduce false positives, support fraud analysts, and enhance the overall security of financial transactions.

3. Objectives

The main objectives of the proposed system are:

- Detect fraudulent financial transactions in real time.
- Improve fraud detection accuracy using Generative AI and LLMs.
- Reduce false positives and false negatives.
- Analyze both structured and unstructured financial data.
- Use GPT to explain suspicious transactions in human-readable language.
- Use BERT for text classification and fraud-related message analysis.
- Apply RLHF to continuously improve the model using analyst feedback.
- Utilize LoRA and PEFT for efficient fine-tuning with lower computational cost.

4. Scope of the Study

- The proposed fraud detection system is designed for various financial domains, including banks, digital payment platforms, credit card companies, insurance organizations, and e-commerce businesses. It monitors financial transactions in real time, identifies suspicious activities, and generates fraud risk scores. The system can process transaction records, customer profiles, login histories, device information, transaction descriptions, and customer complaints to detect fraudulent behavior.
- The framework integrates Large Language Models with machine learning and behavioral analytics, enabling organizations to detect both known and emerging fraud patterns. It also supports continuous learning through analyst feedback, making the system more accurate over time. The proposed solution can be integrated with existing banking systems and payment gateways to strengthen fraud prevention and improve operational efficiency.

5. Importance of the Study

Financial fraud causes billions of dollars in losses every year and affects both organizations and customers. Detecting fraud quickly is essential to reduce financial damage, protect sensitive customer information, and maintain trust in digital financial services.

The proposed AI-based system offers several benefits:

- Detects fraud more accurately than traditional rule-based systems.
- Reduces investigation time by automating fraud analysis.
- Improves customer experience by minimizing false transaction blocks.
- Provides explainable AI-based decisions for analysts.
- Continuously adapts to new fraud techniques through RLHF.
- Supports scalable deployment across different financial institutions.

As digital transactions continue to increase, AI-powered fraud detection systems will become essential for ensuring secure and reliable financial services.

6. Existing System

The existing fraud detection system mainly depends on rule-based algorithms and traditional machine learning models. Financial institutions define a set of rules based on historical fraud patterns. For example, transactions exceeding a specific amount, multiple transactions within a short period, or transactions from unusual locations are flagged for further investigation.

Working of Existing System

1. Customer performs a financial transaction.
2. Transaction data is sent to the fraud detection system.
3. Predefined rules are applied.
4. Machine learning model calculates fraud probability.
5. Transaction is either approved or blocked.
6. Suspicious transactions are forwarded for manual verification.

7.Limitations of Existing System

- Cannot detect newly emerging fraud techniques.
- Generates a high number of false positives.
- Requires frequent rule updates.
- Limited ability to analyze text-based financial data.
- High dependency on manual investigation.

8. Proposed System

- The proposed system combines Generative AI, Large Language Models (LLMs), Natural Language Processing (NLP), and Behavioral Analytics to build an intelligent fraud detection platform.
- Unlike traditional systems, the proposed framework not only analyzes structured transaction data but also understands unstructured information such as transaction descriptions, customer complaints, emails, SMS messages, and fraud investigation reports.

9. Comparison Between Existing and Proposed System

Feature	Existing System	Proposed AI System
Detection Method	Rule-Based	Generative AI + LLM
Learning Ability	Limited	Continuous Learning
Fraud Detection	Known Patterns Only	Known + Unknown Patterns
Data Type	Structured Data	Structured + Unstructured Data
Fraud Explanation	Not Available	AI-Based Explanation
Accuracy	Medium	High
False Positives	High	Low
Adaptability	Low	High
Analyst Support	Manual	AI-Assisted
Scalability	Moderate	Excellent

10. Why Generative AI for Fraud Detection?

Generative AI is capable of understanding complex financial information, recognizing hidden fraud patterns, and generating meaningful explanations for suspicious transactions. Unlike conventional AI models, Generative AI can process both numerical and textual information simultaneously.

Benefits

- Detects complex fraud scenarios.
- Understands customer behavior.
- Generates fraud investigation reports.
- Supports fraud analysts.

11. Model Selection and Justification

To build an intelligent fraud detection system, multiple AI models are integrated, with each model performing a specific task.

11.1 GPT (Generative Pre-trained Transformer)

Purpose

GPT acts as the reasoning engine of the fraud detection system.

Applications

- Fraud explanation
- Suspicious transaction analysis
- Customer complaint summarization

Justification

GPT understands transaction context, identifies suspicious activities, and generates human-readable explanations that help fraud analysts make informed decisions.

Advantages

- Excellent language understanding.
- Strong reasoning capability.
- Generates explainable outputs.

11.2 BERT (Bidirectional Encoder Representations from Transformers)

Purpose

BERT is used for text classification and natural language understanding.

Applications

- Transaction description analysis
- Fraud email detection

Justification

BERT understands the context of financial text more effectively than traditional NLP models, making it suitable for fraud-related text analysis.

Advantages

- High text classification accuracy.
- Strong contextual understanding.
- Suitable for financial documents.

11.3 RLHF (Reinforcement Learning from Human Feedback)

Purpose

RLHF improves model performance using expert feedback from fraud analysts.

Working

1. AI predicts fraud.
2. Fraud analyst reviews prediction.
3. Analyst provides feedback.
4. Model learns from corrections.

Advantages

- Continuous learning.
- Improved decision quality.
- Reduced prediction errors.
- Better adaptation to evolving fraud patterns.

11.4 LoRA (Low-Rank Adaptation)

Purpose

LoRA enables efficient fine-tuning of GPT models using financial datasets.

Why LoRA?

Training a complete LLM requires significant computational resources. LoRA updates only a small number of parameters while keeping the original model unchanged.

Advantages

- Faster training.
- Lower GPU memory usage.
- Reduced computational cost.
- Easy deployment.

11.5 PEFT (Parameter-Efficient Fine-Tuning)

Purpose

PEFT fine-tunes pretrained models by updating only a small subset of parameters.

Benefits

- Lower storage requirements.
- Faster adaptation.
- Efficient deployment.
- Suitable for financial organizations with limited hardware resources.
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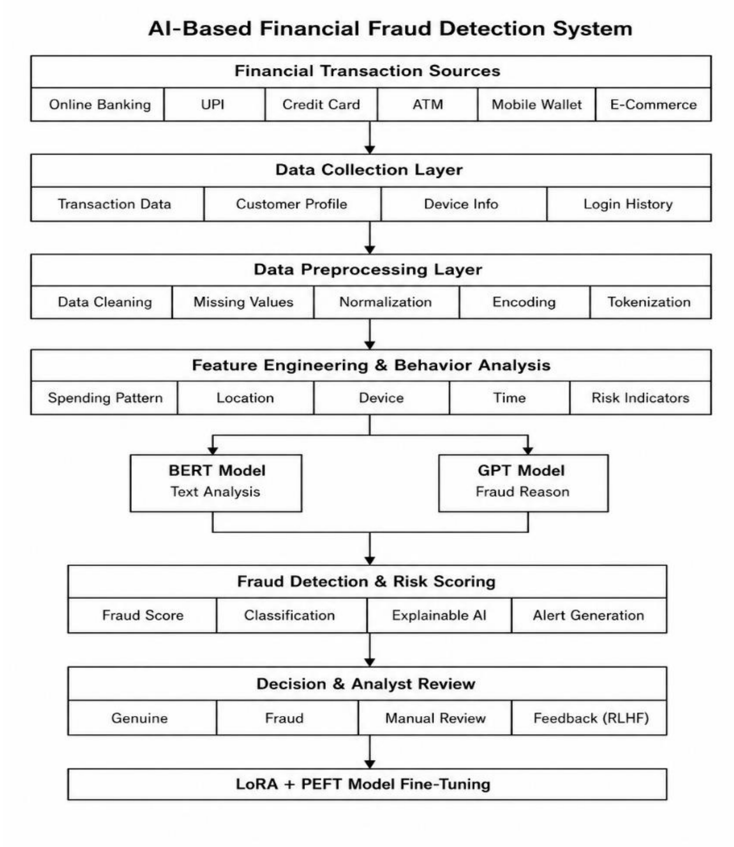
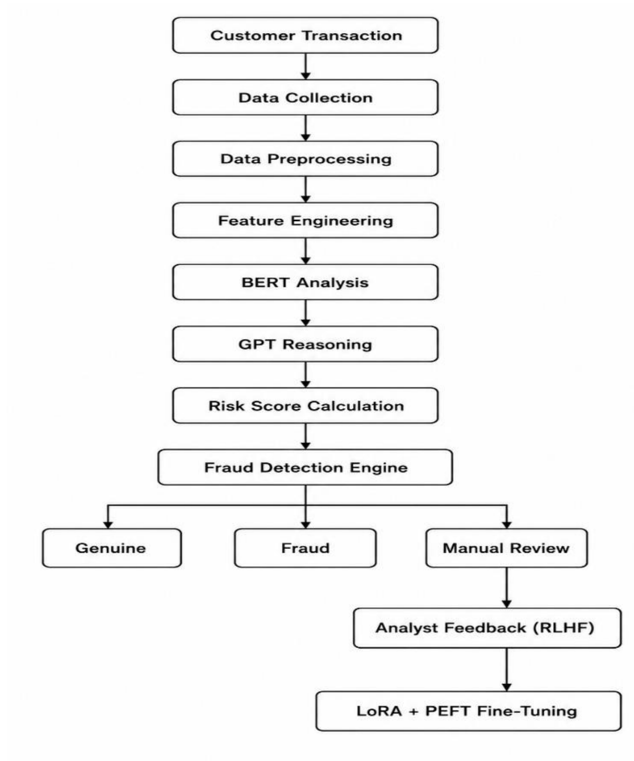
12. Why Combine GPT, BERT, RLHF, LoRA, and PEFT?

Model	**Role in Fraud Detection**
GPT	Fraud reasoning and explanation
BERT	Transaction text classification
RLHF	Continuous improvement using analyst feedback
LoRA	Efficient fine-tuning of GPT
PEFT	Low-cost adaptation of pretrained models

13. System Architecture



14. Workflow of the Proposed System



15. Technologies Used

Component	Technology
Programming Language	Python
Deep Learning	PyTorch / TensorFlow
LLM	GPT
NLP	BERT
Fine-Tuning	LoRA
Efficient Fine-Tuning	PEFT
Feedback Learning	RLHF
Database	PostgreSQL / MySQL
API Framework	FastAPI
Dashboard	Streamlit / Power BI
Cloud Platform	AWS / Azure / Google Cloud

16. Implementation Methodology

The implementation process consists of the following phases:

Phase 1: Data Collection

Gather transaction data from banks, payment gateways, and digital platforms.

Phase 2: Data Preprocessing

Clean and prepare the collected data for analysis.

Phase 3: Feature Engineering

Extract relevant behavioral and transaction features.

Phase 4: Model Training

- Train BERT for text classification.
- Fine-tune GPT using LoRA and PEFT.
- Use RLHF for continuous improvement.

Phase 5: Fraud Detection

Generate fraud scores and classify transactions.

Phase 6: Deployment

Deploy the trained model using cloud services or banking infrastructure.

Phase 7: Monitoring

Monitor system performance and retrain models periodically using analyst feedback.

17. Expected Outcomes

The proposed AI-powered financial fraud detection system is expected to significantly improve the efficiency and accuracy of fraud detection compared to traditional rule-based systems. By integrating Generative AI, Large Language Models (LLMs), and machine learning techniques, the system can identify suspicious transactions in real time while reducing unnecessary alerts.

Expected Outcomes

- Detect fraudulent transactions with high accuracy.
- Reduce false positives and false negatives.
- Provide real-time fraud alerts.
- Generate explainable AI-based fraud reports.
- Improve customer trust and satisfaction.
- Minimize financial losses.

18. Results and Performance Evaluation

- The system performance can be measured using the following key evaluation metrics:

Metric	Purpose
Accuracy	Overall prediction accuracy across all transactions
Precision	Proportion of correctly identified fraud predictions among all flagged alerts
Recall	Proportion of actual fraud cases successfully detected
F1-Score	Harmonic mean balancing precision and recall performance
ROC-AUC	Overall ability of the model to distinguish between fraud and genuine cases

Expected Results

Parameter	Traditional System	Proposed AI System
Accuracy	88–92%	96–99%
Detection Speed	Moderate	Real-Time
False Positives	High	Low
Explainability	Limited	High

19. Advantages

- Detects known and unknown fraud.
- Real-time fraud detection.
- High accuracy with low false positives.
- Explainable AI-based decisions.
- Continuous learning using RLHF.
- Efficient fine-tuning with LoRA and PEFT.
- Improves customer trust.

20. Limitations

- Requires large training datasets.
- High computational resources.
- Periodic model retraining is needed.
- Privacy and security concerns.
- Performance depends on data quality.

21. Future Enhancements

- Integrate Graph Neural Networks (GNNs).
- Use Explainable AI (XAI).
- Blockchain for secure transactions.
- Federated Learning for privacy.
- Multimodal fraud detection.
- Real-time streaming analytics.

22. Conclusion

Financial fraud has become one of the most significant challenges in the digital banking and financial services sector due to the rapid growth of online transactions, digital payment platforms, mobile banking, and e-commerce. Traditional fraud detection methods, which mainly rely on predefined rules and historical transaction patterns, are no longer sufficient to detect evolving and sophisticated fraud techniques.

The proposed AI-powered Financial Fraud Detection System addresses these challenges by integrating Generative AI and Large Language Models (LLMs) with advanced machine learning techniques. GPT is used for intelligent reasoning, fraud explanation, and report generation, while BERT performs accurate classification of transaction-related text and customer communications. RLHF (Reinforcement Learning from Human Feedback) enables the model to continuously improve based on fraud analyst feedback, ensuring that the system adapts to emerging fraud patterns. Additionally, LoRA and PEFT provide efficient fine-tuning, reducing computational cost and making the deployment of large language models more practical for financial institutions.

23. References

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